

Satyam Patel

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SUMMARY

Python & AI/ML Developer skilled in computer vision, agentic AI, FastAPI, and scalable ML deployment. Passionate about optimizing models and building practical, real-world AI solutions.

WORK EXPERIENCE

Research Intern — AI & Computer Vision Jan 2025 – June 2025
Indian Institute of Technology (IIT) BHU [Internship Certificate](#)

- Optimized UAV–IRS pairing using a Genetic Algorithm, improving energy efficiency by **40%**.
- Developed an AI-powered BPPV therapy guide with **real-time head pose tracking**, voice feedback, and error correction.

PROJECTS

AI-Based BPPV Maneuver Guider [GitHub Link](#)

- Developed an **AI-powered** therapy guidance system using **MediaPipe**, **OpenCV**, and **Kalman Filters** for **real-time head pose estimation** (yaw, pitch, roll).
- Integrated **voice-assisted feedback**, error detection, and pose validation with timers and shake recognition to improve accuracy and patient usability.

Meditrack - Patient Management System [Demo](#)

- Built a full-stack patient management application using **React**, **TypeScript**, and **FastAPI** with **Firebase**, featuring complete CRUD operations, real-time search, automatic BMI calculation, and responsive UI with **Tailwind CSS**.
- Deployed frontend on **GitHub Pages** and backend on **Vercel**, implementing RESTful API endpoints for patient data operations with form validation and cross-device accessibility.

EDUCATION

2022 - present	B.Tech(Hons) CSE(Data Science) at UTD-CSVTU,Bhilai	(GPA: 8.3/10.0)
2021	Class 12th CBSE Board	71.2%
2019	Class 10th CBSE Board	70%

SKILLS

Programming Languages	Python, C++, TypeScript, SQL
AI/ML & Computer Vision	MediaPipe, OpenCV, Kalman Filters, Pose Estimation
Web Development	React.js, FastAPI, Node.js, Vite
Cloud & Deployment	GitHub Pages, Vercel, Firebase, RESTful APIs
Development Tools	Git, npm, Responsive Design, CRUD Operations
Other Skills	Data Structures & Algorithms, Problem Solving, Voice Integration

ACHIEVEMENTS

- LeetCode:** Solved 400+ problems with 65% acceptance rate — Contest Rating: 1465 — [Profile](#)